CO3_AT3_MCQ Test

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CO3_AT3_MCQ Test

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The respondent's email (**dhilipk5013.sse@saveetha.com**) was recorded on submission of this form.

1.

A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required? *1/1

- a) 5 TB
- b) 10 TB
- c) 15 TB ✓ (selected)
- d) 20 TB

2.

A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month? *1/1

- a) ~4 minutes ✓ (selected)
- b) ~40 minutes
- c) ~4 hours
- d) ~24 hours

3.

✓ A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second? *1/1

- a)5000
- b) 6000 ✓
- c) 7000
- d) 10,000

4.

✓ A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take? *1/1

- a) 1 hour
- b) 10 hours
- c) 1.1 hours ✓
- d) 5.5 hours

5.

✓ A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year? *1/1

- a) G~4 hours ✓
- b) ~8 hours
- c) ~20 hours
- d) ~40 hours

6.

✓ A cloud provider offers pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost? *1/1

- a) $50
- b) $500 ✓
- c) $1000
- d) $2000

7.

✓ A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput? *1/1

- a) 9.8 Gbps ✓
- b) 9.6 Gbps
- c) 8.0 Gbps
- d) 7.5 Gbps

8.

✓ A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)? *1/1

- a) 91.25 GB ✓
- b) 85 GB
- c)100 GB
- d) 120 GB

9.

A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth? *.../1

- (•) a) 500 Mbps ✗
- ( ) b) 250 Mbps
- ( ) c) 200 Mbps
- ( ) d) 1000 Mbps

No correct answers

10.

A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year? *1/1

- (•) a) 262 minutes ✓
- ( ) b) 365 minutes
- ( ) c) 525 minutes
- ( ) d) 150 minutes

11.

A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center? *1/1

- ( ) a) 4,000
- (•) b) 8,000 ✓
- ( ) c) 6,000
- ( ) d) 10,000

12.

✓ A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted? *1/1

- (●) a) 4 GB ✓
- ( ) b) 6 GB
- ( ) c) -8 GB
- ( ) d) 10 GB

13.

✗ A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted? *0/1

- ( ) a) 16
- ( ) b) 32
- (●) c) 64 ✗
- ( ) d) 128

Correct answer

- (●) b) 32

14.

✓ A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption? *1/1

- (●) a) 40 kW ✓
- ( ) b) 50 kW
- ( ) c) 60 kW
- ( ) d) 70 kW

15.

✓ A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated? *1/1

- a) 6,400
- b) 12,800
- c) 25,600 ✓
- d) 50,000

16.

✓ Which cloud deployment model is used for organizations that require both private and public cloud resources? *1/1

- A) Hybrid Cloud ✓
- B) Public Cloud
- C) Private Cloud
- D) Community Cloud

17.

✓ Which cloud security strategy ensures that sensitive data remains private even if intercepted? *1/1

- A) Encryption ✓
- B) Load balancing
- C) Virtualization
- D) Resource pooling

18.

If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week. *1/1

- a) 5000 GB
- b) 7000 GB
- c) 6000 GB
- d) 7500 GB

19.

A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power? *1/1

- a 56 GHz
- b 58 GHz
- c 64 GHz
- d 70 GHz

20.

If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests. *1/1

- a 1.5 sec
- b 2 sec
- c 1 sec
- d 3 sec

21.

If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month. *1/1

- (•) a 600 GB ✓
- ( ) b 500 GB
- ( ) c 700 GB
- ( ) d 800 GB

22.

A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)? *1/1

- (•) a 21.6 min ✓
- ( ) b 36 min
- ( ) c 43.2 min
- ( ) d 18 min

23.

If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute. *1/1

- (•) a 120 MB ✓
- ( ) b 100 MB
- ( ) c 150 MB
- ( ) d 200 MB

24.

A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement. *1/1

- [x] a 180
- [ ] b 150
- [ ] c 120
- [ ] d 160

25.

A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total? *1/1

- [x] a 1200
- [ ] b 1000
- [ ] c 800
- [ ] d 600

26.

If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently? *1/1

- [x] a 12 min
- [ ] b 10 min
- [ ] c 6 min
- [ ] d 8 min

27.

✓ A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day. *1/1

Formula: Energy = Units × Power × Time (hrs)

- (◉) a 420 kWh ✓
- ( ) b 480 kWh
- ( ) c 400 kWh
- ( ) d 500 kWh

28.

✓ Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy? *1/1

- (◉) a 32 TB ✓
- ( ) b 35 TB
- ( ) c 30 TB
- ( ) d 28 TB